

# Phil R. Van-Lane

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## Profile

I am a 4th year PhD candidate in the David A. Dunlap Department of Astronomy & Astrophysics at University of Toronto, and also currently a Visiting Graduate Student at University of California San Diego. My research is focused on developing statistical and machine learning methodologies to improve age constraints for low mass stars, and applying those constraints to study exoplanet evolution. I am actively engaged within the U of T department and community, having served as Co-President of the Graduate Astronomy Students Association in 2023-2024 and in many other roles. I also work as an instructor for DSI Certificates (a program designed to rapidly upskill people targeting a career in data science and machine learning). Before beginning my PhD, I worked for seven years in data engineering and software development.

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## Education

### University of Toronto

PhD, Astronomy & Astrophysics

Toronto, ON, Canada

2022-2027 (expected)

Advisors: Prof. Joshua Speagle, Prof. Gwendolyn Eadie, Prof. Ryan Cloutier (McMaster University)

UCSD Host: Prof. Chris Theissen

Thesis Title: *From Stars to Statistics: Examining Exoplanet Populations through Stellar Activity with Machine Learning.*

### University of Waterloo

BSc, Earth Science (Geophysics specialization)

Waterloo, ON, Canada

2010-2015

Advisor: Prof. Tony Endres

Thesis Title: *An Examination of GPR Processing and its Effect on Data Interpretation.*

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## Research

### Refereed Publications

1. **Van-Lane, P. R.** et al., 2025, "ChronoFlow: A Data-Driven Model for Gyrochronology", *ApJ*, 986, 59, <http://doi.org/10.3847/1538-4357/adcd73>.
2. Khandelwal, A., et al. (9 co-authors incl. **Van-Lane, P.**), 2024, "Beyond CCDs: characterization of sCMOS detectors for optical astronomy" (proceedings from the 2024 SPIE Astronomical Telescopes + Instrumentation Workshop), <https://arxiv.org/abs/2409.16449>.
3. **Van-Lane, P.**, Speagle, J. S., & Douglas, S. 2023, "A Novel Application of Conditional Normalizing Flows: Stellar Age Inference with Gyrochronology" (proceedings from the 2023 ICML Workshop on Machine Learning for Astrophysics), <https://arxiv.org/abs/2307.08753>.

### Presented Talks and Posters

1. Contributed poster, [Astronomical Data Analysis Software & Systems](#) November 2025
2. Contributed talk, [Joint Statistical Meetings](#) August 2025
3. Contributed talk, [9th TESS/16th Kepler Asteroseismic Science Consortium](#) July 2025

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|------------------------------------------------------------------------------------------------------------------------|--------------|
| 4. <u>Contributed talk</u> , <a href="#">246th AAS Meeting</a>                                                         | June 2025    |
| 5. <u>Contributed talk</u> , <a href="#">Statistical Society of Canada Annual Meeting</a>                              | May 2025     |
| 6. <u>Invited talk</u> , <a href="#">UCSD "SMASH" Series</a>                                                           | March 2025   |
| 7. <u>Invited talk</u> , <a href="#">CfA Seminars</a>                                                                  | October 2024 |
| 8. <u>Contributed talk</u> , <a href="#">8th TESS/15th Kepler Asteroseismic Science Consortium</a>                     | July 2024    |
| 9. <u>Contributed poster</u> , <a href="#">The 22nd Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun</a> | June 2024    |
| 10. <u>Contributed poster</u> , <a href="#">CASCA AGM</a>                                                              | June 2024    |
| 11. <u>Contributed talk</u> , <a href="#">Globular Clusters and Their Tidal Tails</a>                                  | May 2024     |
| 12. <u>Contributed talk</u> , <a href="#">Great Lakes Clusters and Streams</a>                                         | August 2023  |
| 13. <u>Contributed talk</u> , <a href="#">7th TESS/14th Kepler Asteroseismic Science Consortium</a>                    | July 2023    |
| 14. <u>Contributed poster</u> , <a href="#">ICML Workshop on Machine Learning for Astrophysics</a>                     | July 2023    |
| 15. <u>Contributed talk</u> , <a href="#">CASCA AGM</a>                                                                | June 2023    |

## Scholarships and Funding

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### Doctoral Scholarships and Fellowships (\$175,500 total funding awarded)

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|----------------------------------------------------------------|-----------|
| NSERC Postgraduate Scholarship (PGS-D); \$120,000 over 3 years | 2024-2027 |
| Walter C. Sumner Memorial Fellowship; \$9,000                  | 2025      |
| Lachlan Gilchrist Fellowship Fund; \$4,000                     | 2024-2025 |
| QEII Graduate Scholarship in Science and Technology; \$15,000  | 2023-2024 |
| Robert Shelley Graduate Scholarship; \$5,000                   | 2023-2024 |
| NSERC Graduate Scholarship (CGS-M); \$17,500                   | 2022-2023 |
| Department of Astronomy & Astrophysics Entrance Award; \$5,000 | 2022-2023 |

### Other Doctoral Awards and Recognitions

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|----------------------------------------------------------------------|------|
| Winner, Student Research Oral Presentation Award, SSC Annual Meeting | 2025 |
| Finalist, Astrostatistics Interest Group Student Paper Competition   | 2025 |
| 3rd Place, CoolStars22 Early Career Researcher Poster Competition    | 2024 |
| "You Lead, We Follow" Dunlap Institute Award                         | 2024 |

### Undergraduate Scholarships and Awards

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|-----------------------------------------------------------------------|-----------|
| Dean's Honours List (every term)                                      | 2010-2015 |
| Logan Student Prize                                                   | 2015      |
| Suncor Emerging Leaders Award in Science                              | 2011-2015 |
| J.P. Bickell Foundation Scholarship                                   | 2011-2015 |
| Queen Elizabeth II Aiming for the Top Scholarship                     | 2011-2015 |
| Brantford Lapidary and Mineral Society Inc. Scholarship               | 2012-2013 |
| UW President's Research Award                                         | 2012      |
| NSERC Undergraduate Research Award                                    | 2012      |
| Phillip Hall of Memorial Scholarship                                  | 2012      |
| Canadian Exploration Geophysical Society Foundation Scholarship Award | 2012      |
| UW President's Scholarship of Distinction                             | 2010-2011 |
| Conestoga Rover's and Associates Entrance Scholarship                 | 2010-2011 |

## Service and Academic Leadership

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### Department Leadership

*\*DADDAA refers to roles held with the David A. Dunlap Department of Astronomy & Astrophysics*

*\*GASA refers to roles held with the Graduate Astronomy Students' Association at U of T*

**Co-President, GASA**

2023-2024

- Negotiated a  $\approx 10\%$  raise to the guaranteed funding package for graduate students, and an increased top-up for major awards.
- Managed the activities of all GASA committees, coordinated prospective student visits, and aligned GASA feedback for a faculty search.

#### **Training and Mentoring Committee, DADDAA**

2022-2023

- Organized and coordinated training and mentorship activities for all department members.

#### **Mediation Committee, GASA (x2)**

2022-2025

- Developed and conducted a survey on financial health of GASA members.
- Used the survey results to provide recommendations to the department for funding adjustments.

#### **Other GASA Positions**

Web Director, GASA

2025-2026

Graduate Student Mentor (x2), GASA

2023-2025

Social Committee (x2), GASA

2022-2025

Graduate Student Mentorship Coordinator, GASA

2022-2023

### **Conference and Workshop Organization**

#### **Organizing Committee, STATSTRO Workshop**

2024-2025

- Helped to coordinate content and scheduling for both 2024 and 2025 versions of this workshop designed to bring astronomers and statisticians together.
- Designed and developed the [STATSTRO 2025](#) website.

### **Teaching**

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#### **Head TA Positions (UofT)**

AST201: Stars and Galaxies, Head Tutorial TA

2024

- Coordinated duties for graduate and undergraduate TAs across the entire course.
- Ensured feedback and evaluation was consistent from all TAs for projects and testing.

AST101: The Sun and Its Neighbours, Head TA

2024

- Designed tutorial content and structure for  $\approx 30$  graduate TAs to deliver to  $\approx 1500$  undergraduate students over the duration of the course.
- Ran weekly meetings with all of the TAs to make sure everyone was aligned on the course content and delivery.

#### **Other Graduate TA Positions (UofT)**

AST201: Stars and Galaxies (x3)

2023-2025

AST101: The Sun and Its Neighbours (x2)

2022-2023

#### **Undergraduate TA Positions (UW)**

EARTH460: Applied Geophysics II

2015

EARTH260: Applied Geophysics I

2013

EARTH359: Flow Through Porous Media

2012

### **Community Engagement and Outreach**

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#### **Team Project Instructor, DSI Certificates**

2024-2025

- This is a program designed to rapidly upskill individuals (with an emphasis on recent immigrants) who have employment experience but are looking to transition into a machine learning or data science career.
- In my 2-week module (which I have delivered for the last 3 cohorts of students), I focus on developing the participants' "softer" skills required to excel in a technical career, such as self-organizing teamwork and delivering clear value to stakeholders.

- Volunteer Coordinator, AstroTours** 2022-2023
- Coordinate ≈20 volunteers for this monthly outreach event which includes a public talk and interactive astronomy activities.
- Volunteer, Astronomy On Tap Toronto** 2022-2024
- Engaged members of the public and assisted with logistics for these events which are the largest astronomy outreach events in Toronto.
- Astronomy Educator, British Columbia School District #36** 2024
- Presented an introductory astronomy classroom talk to Grade 6 students.
- Volunteer, AstroEDU Toronto** 2023
- Helped facilitate this conference designed to address common issues in astronomy education and share experiences among astronomy educators.

## Previous Employment

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### Hootsuite Inc.

- Product Manager, Data Technology** 2021 - 2022
- Developed the product roadmap for the company's entire data engineering department.
  - Managed an initiative to standardize all of the company's data pipelines.

### Royal Bank of Canada

- Senior Product Manager, Data Innovation Lab** 2020 - 2021
- Led the technical implementation of the reporting tool for RBC's global "Return To Premises" (post-COVID) initiative.
  - Developed the product roadmap for my team's in-house data visualization application.
- Product Manager, Data Innovation Lab** 2019 - 2020
- Expanded the reach of my team's internal data visualization product into several departments across the company.
- Senior Developer, Data Innovation Lab** 2018 - 2019
- Full stack developer.
  - Built the the first web application to be deployed on RBC's internal cloud.
- Technical Systems Analyst, Corporate Real Estate IT** 2016 - 2018
- Developed and automated ≈20 data pipelines for the Corporate Real Estate team.
  - Deployed software to devices (eg. self-service kiosks, meeting room booking panels) throughout RBC office buildings.

### Tetra Tech EBA

- Geotechnician** 2016
- Conducted materials testing and quality control for various construction projects at remote sites throughout Northwest Territories and Nunavut.
  - Configured a remote alerting system for seismic vibrations induced during construction work at a hospital.

### Co-op Jobs (while at UW)

- Geophysicist, Nexen Energy ULC** Jan-Aug 2014
- Geotechnician, Tetra Tech EBA** May-Sep 2013
- Researcher, UW Geophysics Lab** Sep-Dec 2012
- Geologist, Lake Shore Gold Corp.** Jan-Apr 2012
- Geologist, Lake Shore Gold Corp.** May-Aug 2011